S8-CFD: A Lightweight From-Scratch Context-Fused Detector for Small Drone Detection

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***Abstract*—This report presents S8-CFD, a lightweight single-class drone detector designed and implemented from scratch in PyTorch. The dataset and problem setting emphasize small-object localization: all source images are 2560×1440, contain annotated drones, and are grouped to reduce leakage from related sequential and raw/augmented samples. S8-CFD uses a stride-8 anchor-free prediction grid with S8/S16/S32 context fusion, GroupNorm, custom decoding, and custom non-maximum suppression. The final baseline obtains validation AP50/AP75/mAP50-95 of 0.9376/0.4038/0.4792 and sealed-test AP50/AP75/mAP50-95 of 0.8811/0.3026/0.4115. At the validation-selected threshold of 0.999, sealed-test precision, recall, and F1 are 0.9437, 0.8584, and 0.8990. A standard IoU-loss ablation improves validation AP50 and operating F1 slightly, but does not improve AP75 or the predetermined validation mAP50-95 selection metric.**

I. Introduction

Small drones are visually compact and often appear against sky, buildings, or natural scenery. This makes coarse detection easier than precise localization: the frozen baseline has high AP50 but much lower AP75, motivating a careful localization-focused analysis. The implementation is intentionally developed from scratch without pretrained detector weights, YOLO/Ultralytics, torchvision detector models, timm backbones, or imported detection architectures.

The resulting model, S8-CFD, is a 3,242,589-parameter anchor-free detector. It uses stride-8 small-object prediction and fuses S8, S16, and S32 feature context in a custom head. The implementation includes from-scratch target encoding, decoding, IoU computation, and NMS. PyTorch is used as the tensor and training framework [1]; GroupNorm follows the normalization choice of Wu and He [2]. Multi-scale context fusion is related in spirit to feature pyramid reasoning [3], but the architecture here is not an imported FPN detector.

II. Dataset and Evaluation Protocol

The dataset contains 2400 source images and 4800 annotated objects. Each image is 2560×1440 and the annotation set has one semantic class: drone. Although the dataset contains two annotated drones per image, the detector and inference code do not hard-code two detections; inference emits a variable number of post-NMS predictions.

The split is grouped to reduce leakage from related samples. A filename-derived `capture_pair` field is used as an inferred proxy for source identity or sequential relation, not as ground-truth identity. Train, validation, and test groups are separated to reduce overlap from sequentially related samples and raw/augmented variants. Table I summarizes the frozen split.

TABLE I
Frozen dataset split.

Split	Images	Objects
Train	1696	3392
Validation	245	490
Test	459	918
Total	2400	4800

The test set remained sealed until the final model, checkpoint, and operating threshold were frozen. The final model is the validation-best S8-CFD baseline at epoch 19, selected by validation mAP50-95. AP is computed from ranked candidates retained at `EVAL_CONFIDENCE_FLOOR`=0.001, following the standard ranked detection-evaluation convention used in COCO-style AP analysis [4]. Precision, recall, and F1 are computed at an operating confidence threshold. The final threshold 0.999 was selected on validation only. NMS IoU is 0.50, `EVAL_TOP_K`=500, and `MAX_DETECTIONS`=100. The sealed test was evaluated once after freezing the model and threshold.

III. S8-CFD Detector

Input images are resized to 960×540 and vertically padded to a 960×544 canvas. S8-CFD predicts on a stride-8 68×120 grid. Each cell outputs five values: an objectness logit, x/y cell offsets, and log width/height. Positive cells are assigned by object centers.

The model uses custom convolutional blocks with GroupNorm and context fusion from S8, S16, and S32 feature levels. The detection head is anchor-free: decoded centers use sigmoid-normalized cell offsets, sizes use exponentiated log width/height, and boxes are converted to normalized `xyxy` coordinates. NMS is implemented in PyTorch tensor operations rather than through an external library. Figure 1 summarizes the implemented architecture.

The baseline objective is

$$L_{\mathrm{base}} = L_{\mathrm{obj}} + 5L_{\mathrm{center}} + 2L_{\mathrm{size}}, \tag{1}$$

where objectness is balanced binary cross entropy over positive and negative cells, and center and size terms use Smooth L1. Training uses AdamW [5].

The localization ablation adds a standard IoU term for positive assigned cells:

$$L_{\mathrm{exp}} = L_{\mathrm{base}} + \lambda_{\mathrm{iou}} L_{\mathrm{iou}}, \quad \lambda_{\mathrm{iou}} = 1, \tag{2}$$

$$L_{\mathrm{iou}} = 1 - \mathrm{IoU}. \tag{3}$$

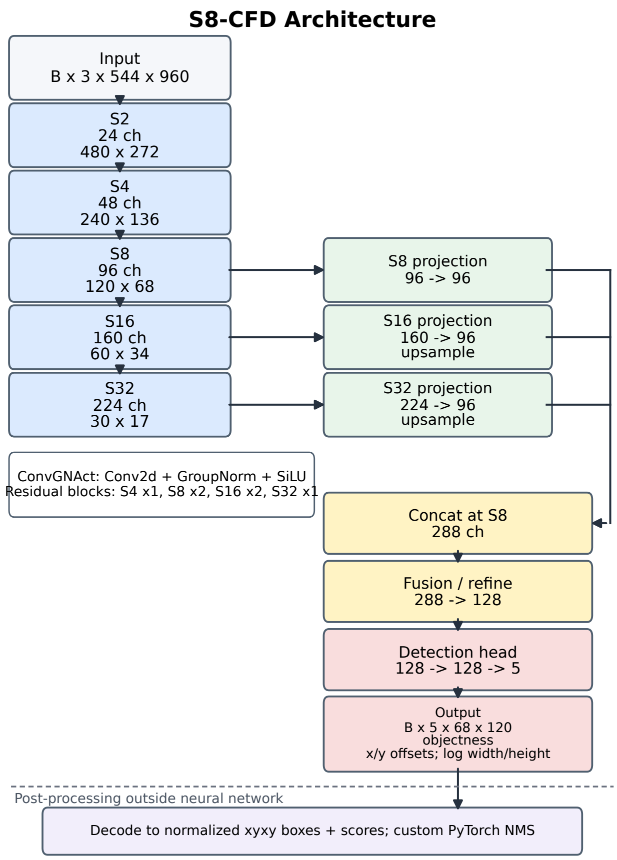

Fig. 1. S8-CFD architecture. Features from strides 8, 16, and 32 are projected and fused at stride 8 before the anchor-free detection head.

No GIoU, DIoU, or CIoU variant is used in this ablation.

IV. Experiments and Results

The reference baseline configuration is fixed for 20 epochs, batch size 2, learning rate 0.001, weight decay 0.0001, no learning-rate scheduler, and seed 42. Training used Windows CUDA on an NVIDIA GTX 1660 SUPER with `NUM_WORKERS`=4. W&B tracking was used for experiment logging, while the repository also supports CPU Docker smoke runs and platform-aware CUDA/MPS/CPU local execution.

Table II gives the compact validation and final sealed-test summary. For the IoU ablation, AP metrics come from the validation-best checkpoint, while precision, recall, and F1 use the validation-selected threshold from the threshold-sweep artifact.

The IoU ablation is neutral to negative with respect to the predetermined selection metric. It improves validation AP50 from 0.9375629425 to 0.9507038593 and slightly improves operating F1 from 0.9338929696 to 0.9364278507. However, AP75 decreases from 0.4037680626 to 0.4022931159, and mAP50-95 decreases from 0.4791892208 to 0.4783972635. The mAP difference is only about 0.0008, so it should not be overstated; nevertheless, the ablation does not improve

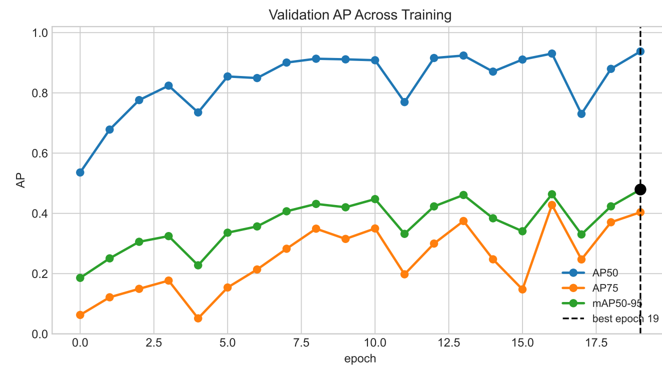

Fig. 2. Baseline validation AP curves. The validation-best checkpoint is selected by mAP50-95, not by operating F1.

the predetermined validation criterion. The baseline therefore remains the final selected model.

The sealed-test evaluation of the frozen baseline contains 459 images and 918 objects. At threshold 0.999 it produces 788 true positives, 47 false positives, and 130 false negatives, with 835 displayed predictions or 1.819 predictions per image. Relative to validation, sealed-test AP50 changes by -0.05645, AP75 by -0.10117, mAP50-95 by -0.06766, and F1 by -0.03486.

V. Discussion and Limitations

The final detector achieves AP50 of 0.8811 and operating precision of 0.9437 on the sealed test split. Performance decreases relative to validation, especially at AP75. This supports the interpretation that stricter localization generalizes less strongly than AP50-level detection. Held-out capture-pair conditions may contribute to the gap, but the evidence does not establish causality and should not be labeled definite overfitting.

Several limitations remain. The dataset is relatively small, single-class, and structured around inferred filename metadata. The split grouping is a practical leakage-reduction proxy rather than verified scene identity. The main training results use one primary seed. The IoU experiment tests only standard IoU with $\lambda_{\text{iou}} = 1$; it does not exhaust localization-loss choices. The results do not imply broad real-world deployment readiness.

VI. Conclusion

S8-CFD demonstrates that a compact, from-scratch, context-fused anchor-free detector can provide useful drone detection under a leakage-aware validation and sealed-test protocol. The final selected baseline achieves sealed-test AP50/AP75/mAP50-95 of 0.8811/0.3026/0.4115 and precision/recall/F1 of 0.9437/0.8584/0.8990 at the frozen validation-selected threshold. A standard IoU-loss ablation did not improve the predetermined validation mAP50-95 criterion, so it serves as useful neutral/negative evidence rather than a new selected model. Implementation, final weights, frozen experiment artifacts, Docker CPU smoke support, and executed notebooks are included in the repository.

TABLE II

MAIN VALIDATION AND SEALED-TEST RESULTS. AP USES RANKED CANDIDATES; P/R/F1 USE EACH LISTED OPERATING THRESHOLD.

Run	Epoch	Threshold	AP50	AP75	mAP	Prec.	Rec.	F1
Baseline validation	19	0.999	0.9376	0.4038	0.4792	0.9611	0.9082	0.9339
IoU $\lambda = 1$ validation	17	0.998	0.9507	0.4023	0.4784	0.9261	0.9469	0.9364
Final sealed test	19	0.999	0.8811	0.3026	0.4115	0.9437	0.8584	0.8990

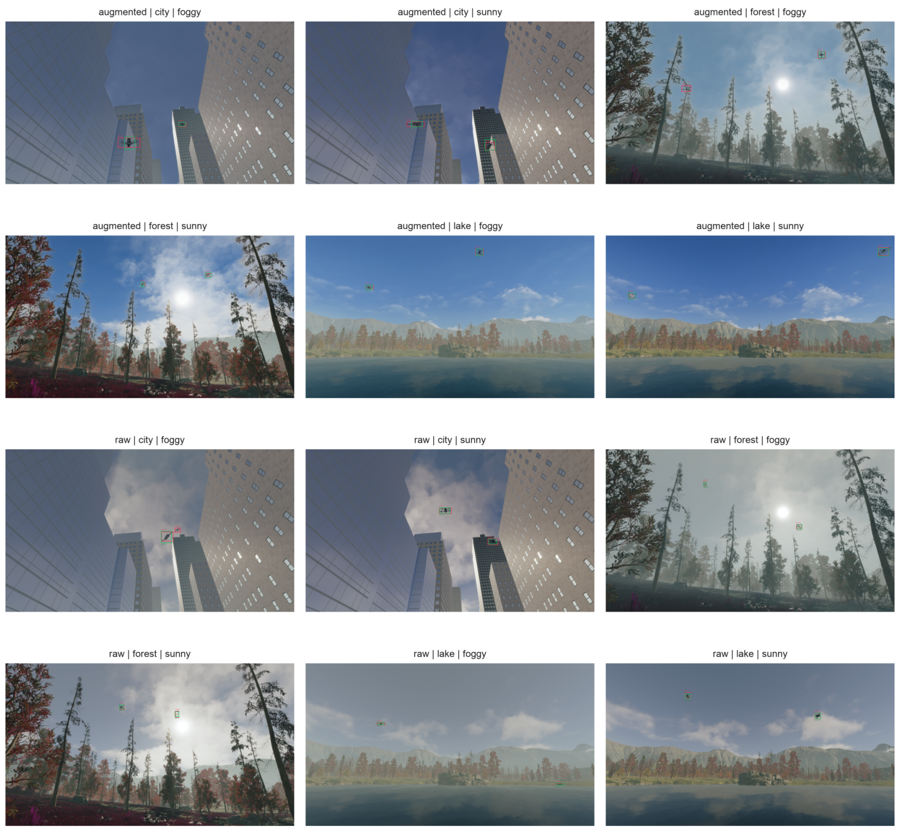

Fig. 3. Representative sealed-test predictions. The 12-image subset was curated for coverage across raw/augmented, city/forest/lake, and sunny/foggy conditions, not for correctness or confidence.

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